ITC 353 Project 1 Worksheet

NYC 311 Service Request Delay Prediction

Name:

Part 1: Business Understanding

1. Business Problem

In your own words, describe the business problem your team is trying to solve.

Response:

[Type your response here.]

Notes : A service request is your request for the City to provide you with assistance, perform an inspection, or address a problem. NYC311 can accept service requests for a wide range of issues, including over 500 complaint types.

Common Examples

- Fixing a neighborhood pothole
- Reporting a lack of heat or hot water in an apartment
- Requesting a refund for an overpaid parking ticket
- Reporting noise disturbances or illegal parking
- Registering rat or pest conditions.

Some Numbers :

1. As measured through NYC311 requests, the demand for city services has surged in recent years. Calls to 311 climbed to over 3.4 million in 2024, up 7% from 2023.
2. **Illegal parking complaints** topped over half a million in 2024, a 155% increase since 2019.
3. **Noise complaints** reached over 610,000 in 2024, a 19% increase from 2023, disproportionately affecting certain communities.
4. **Heat and hot water complaints** rose over 246,700 in 2024, up more than 14% from 2019. The Bronx had more than 192 complaints per 1,000 residents in the Fordham, Bedford Park and Norwood sections.

2. Prediction Goal

What are you trying to predict in this project?

Linear regression model:

[Type your response here.]

Logistic regression model:

[Type your response here.]

3. Business Use

How could city operations leaders use these predictions?

Response:

[Type your response here.]

Part 2: Data Understanding and Exploratory Data Analysis

Before building models, explore the dataset carefully. Your goal in this section is to understand what the data contains, what problems need to be addressed, and which variables may be useful for prediction.

4. Initial Data Review

Record the basic structure of your dataset. You may not be able to complete the bottom half of the table right away. Return to it after you have finished inspecting the data, making cleaning decisions, and selecting the variables you will use for modeling.

Item	Answer
Number of rows before cleaning	40001
Number of columns before cleaning	16
Number of rows after removing unusable records	
Number of columns after selecting/creating variables for modeling	

Briefly explain any major changes in the number of rows or columns.

Response:

[Type your response here.]

5. Missing Values and Data Quality

Create a missing values summary table for all columns. Note: this may be an iterative process. Recommendation: fill in the first three columns first, then complete the rest of the data exploration before making final decisions about removal or recoding.

Variable	Number Missing	Percent Missing	Keep, remove, or recode?	Reason

Based on your missing values summary, answer the following:

Which variables have the most missing data?

[Type your response here.]

Are any variables too incomplete to be useful? Explain.

[Type your response here.]

Are there missing values that could be meaningful? For example, does a blank value sometimes mean "not applicable" rather than an error?

[Type your response here.]

6. Categorical Variable Distributions

Create frequency tables or bar charts for the following variables: agency, problem, problem_detail, location_type, borough, community_board, and open_data_channel_type. For each variable, focus on the most common categories.

Variable	What are the most common categories?	Are there many rare categories?	Modeling decision
agency		Yes / No	
problem		Yes / No	
problem_detail		Yes / No	
location_type		Yes / No	
borough		Yes / No	
community_board		Yes / No	
open_data_channel_type		Yes / No	

Which categorical variables seem most useful for predicting resolution time? Why?

[Type your response here.]

Which categorical variables may be difficult to use because they have too many categories or too many rare values?

[Type your response here.]

Did your team decide to group, remove, or simplify any categorical variables? Explain.

[Type your response here.]

7. Date and Time Patterns

Use created_date to explore when service requests were submitted. Create at least two of the following:

- Requests by day of the month
- Requests by day of week
- Requests by hour of day
- Requests on weekdays versus weekends

Are service requests evenly distributed across the month?

[Type your response here.]

Do requests appear more common on certain days of the week or times of day?

[Type your response here.]

What date/time features did your team create for modeling? Why might they be useful?

[Type your response here.]

8. Resolution Time Distribution

Create a resolution_hours variable using created_date and closed_date. Then create the following:

1. A summary table for resolution_hours
2. A histogram of resolution_hours
3. A histogram of log_resolution_hours

Use this table to record your summary statistics.

Statistic	Value
Minimum	
Mean	
Median	
75th percentile	
90th percentile	
95th percentile	
Maximum	
Number of requests with 0 resolution hours	
Number of requests over 30 days	

What does the histogram of raw resolution_hours show?

[Type your response here.]

Why might log_resolution_hours be useful for the linear regression model?

[Type your response here.]

Are there very short or very long resolution times? How did your team decide to handle them?

[Type your response here.]

9. Resolution Time by Key Variables

Explore how resolution time differs across important categories. Create at least three of the following:

- Median resolution time by agency
- Median resolution time by problem
- Median resolution time by borough
- Median resolution time by open_data_channel_type
- Boxplot of resolution time by agency
- Boxplot of resolution time by borough

Record key findings below.

Variable	What did you compare?	What pattern did you notice?	Why might this matter for modeling?

Which agencies or problem types tend to have longer resolution times?

[Type your response here.]

Which agencies or problem types tend to be resolved quickly?

[Type your response here.]

Why might it be unfair to use one single definition of "slow" for every type of service request?

[Type your response here.]

10. Relationships Between Important Variables

Create at least two cross-tabulations to explore relationships between categorical variables. Recommended cross-tabs: agency by problem, problem by borough, problem by open_data_channel_type, and borough by open_data_channel_type.

Cross-tab	What relationship did you examine?	What did you learn?
1.		
2.		

Are some problem types handled mostly by specific agencies?

[Type your response here.]

Do some types of requests appear more common in certain boroughs or submission channels?

[Type your response here.]

How could these relationships affect your model results or interpretation?

[Type your response here.]

11. EDA Summary

Based on your exploratory analysis, summarize the most important things your team learned before modeling.

Key EDA finding	Why it matters for the project
1.	
2.	
3.	

Which variables do you expect to be most useful for prediction? Explain.

[Type your response here.]

What is one data issue your team will need to address before modeling?

[Type your response here.]

Part 3: Data Preparation

12. Resolution Time

How did your team calculate resolution time?

Formula or explanation:

[Type your response here.]

What unit did you use?

[ ] Minutes [ ] Hours [ ] Days [ ] Other: _____

13. Records Removed

What records did your team remove before modeling?

Records removed	Reason

14. New Variables Created

List the new variables your team created.

New variable	How it was created	Why it may be useful

15. Missing Values

How did your team handle missing values?

Variable or field	Decision	Reason

16. Categorical Variables

Which categorical variables did your team use in the models?

Categorical variable	How you handled it	Reason

Part 4: Data Leakage Check

A model should only use information that would be available when the service request is first submitted.

17. Leakage Variables

Identify any variables that should not be used as predictors because they would create data leakage.

Variable	Why it should not be used as a predictor

18. Final Predictor Set

List the variables your team used as predictors in your final models.

Predictor	Used in linear model?	Used in logistic model?	Reason for including
	Yes / No	Yes / No	
	Yes / No	Yes / No	
	Yes / No	Yes / No	
	Yes / No	Yes / No	
	Yes / No	Yes / No	

Part 5: Holdout Validation

19. Training and Validation Split

In this project, you will use the holdout method to evaluate your models. The holdout method divides the data into two independent parts:

- The training set is used to build the model.
- The validation set is used to evaluate how well the model predicts new, unseen observations.

Record how your team split the data.

Item	Answer
Percent used for training	
Percent used for validation	
Number of observations in the training set	
Number of observations in the validation set	
Random seed used, if applicable	

Why did your team use this split?

[Type your response here.]

Why is it important to evaluate model performance on the validation set instead of only using the training set?

[Type your response here.]

Part 6: Linear Regression Model

20. Linear Regression Outcome

What outcome variable did you use for the linear regression model?

[Type your response here.]

Did you transform the outcome variable?

[ ] No [ ] Yes — explain the transformation:

[Type your response here.]

Why was this outcome appropriate for predicting resolution time?

[Type your response here.]

21. Linear Regression Validation Performance

Build the linear regression model using the training set. Then use the model to predict the outcome for the validation set.

Record your model performance.

Metric	Training Set	Validation Set
RMSE		
MAE, if used		
R-squared, if used		

Use RMSE as your primary performance measure. A lower RMSE means the model's predictions are closer to the actual values.

What does the validation RMSE tell you about the model's predictive performance?

[Type your response here.]

Is there a large difference between the training RMSE and validation RMSE? What might that suggest?

[Type your response here.]

22. Linear Regression Interpretation

What did the linear regression model tell you?

[Type your response here.]

Was the model useful from a business perspective? Why or why not?

[Type your response here.]

Part 7: Logistic Regression Model

23. Defining a Slow Request

For the logistic regression model, the response variable should be binary:

- 1 = slow request, the target class
- 0 = not slow request, the nontarget class

How did your team define a "slow" service request?

[Type your response here.]

Why is this definition reasonable for the business problem?

[Type your response here.]

24. Logistic Regression Validation Performance

Build the logistic regression model using the training set. Then use the model to predict slow requests in the validation set.

What probability cutoff did your team use to classify requests as slow?

[ ] 0.50 [ ] Other cutoff: _____

If you used a cutoff other than 0.50, explain why.

[Type your response here.]

Complete the confusion matrix for your validation set.

	Predicted Not Slow	**Predicted Slow**
Actually Not Slow		
Actually Slow		

Record your classification performance measures.

Metric	**Validation Set Value**

Accuracy	
Sensitivity / Recall for slow requests	
Specificity	

Accuracy measures the overall percentage of correctly classified requests. Sensitivity measures the percentage of actual slow requests that were correctly flagged. Specificity measures the percentage of actual not-slow requests that were correctly identified.

Which performance measure is most important for this business problem? Explain.

[Type your response here.]

25. Logistic Regression Interpretation

What types of requests were more likely to be classified as slow?

[Type your response here.]

Which error is more concerning for this business problem?

[ ] Predicting a request will be slow when it is not slow [ ] Predicting a request will not be slow when it actually is slow

Explain your answer.

[Type your response here.]

Part 8: Business Recommendation

26. Main Findings

Summarize your three most important findings.

Finding	Evidence from your analysis
1.	
2.	
3.	

27. Recommendation

What should city operations leaders do based on your analysis?

[Type your response here.]

28. Responsible Use

What should leaders be careful about if they use this model?

[Type your response here.]

29. Limitations

Identify at least two limitations of your analysis.

Limitation	Why it matters
1.	
2.	

Part 9: AI and Tool Use Reflection

30. Tool Use

Did you use AI or other tools to support your work?

[ ] No [ ] Yes

If yes, briefly describe how you used the tool and how you checked the output.

[Type your response here.]

31. Learning Reflection

What is one data preparation or modeling decision you understand better after completing this project?

[Type your response here.]

Reminder: Your worksheet should explain your reasoning. Do not simply paste code output without interpreting what it means for the project.